

View Meta-Reviews

Paper ID

2635

Paper Title

ClearEval: A Data-driven Benchmark for Evaluating Large Language Models in Tissue Optical Clearing

META-REVIEWER #1

META-REVIEW QUESTIONS

7. Please justify your decision. In case you deviate from the reviewers' recommendations, explain in detail the reasons why. In case of an invitation for rebuttal, clarify which points are important to address in the rebuttal.

The paper itself is indeed well-written and clearly-structured. However, the entire evaluation protocol is questionable as the benchmark is highly dependent on LLM-as-a-Judge.

The authors already investigate the correlation of automated metrics with human annotation and therefore the evaluation itself should still hold effective. However, the reviewers all point out that the lack of related work, the limited comparison with SOTA methods w/ and w/o RAG/tool-use settings can lead to biased conclusions on the capabilities of existing LLMs (especially LLM/LVM-driven agents) in TOC tasks. The authors are encouraged to respond with evidence.

9. [Rebuttal Paper Only] After you have reviewed the rebuttal and updated reviews, please provide your recommendation based on all reviews and the authors' rebuttal.

Reject

10. [Rebuttal Paper Only] Please justify your recommendation. (visible to authors after final paper notification)

Reviewers all question the validity and generalizability of the benchmark, mostly around the limited scope and data curation transparency. The rebuttal did provide useful clarification but fail to resolve most of the concerns. The authors are encouraged to resubmit and prepare further details of data curation for a benchmark paper (e.g., supplementary/appendix materials).

META-REVIEWER #2

META-REVIEW QUESTIONS

9. [Rebuttal Paper Only] After you have reviewed the rebuttal and updated reviews, please provide your recommendation based on all reviews and the authors' rebuttal.

Reject

10. [Rebuttal Paper Only] Please justify your recommendation. (visible to authors after final paper notification)

The reviewers provided mixed post-rebuttal feedback. While they acknowledge the value of the work and the interesting application, some critical issues raised remain, including its contribution when positioned against existing large-scale similar-purposed work such as BioProBench (27k protocols), overclaims of contributions, and a lack of transparency in the key benchmark construction.

META-REVIEWER #3

META-REVIEW QUESTIONS

9. [Rebuttal Paper Only] After you have reviewed the rebuttal and updated reviews, please provide your recommendation based on all reviews and the authors' rebuttal.
Reject

10. [Rebuttal Paper Only] Please justify your recommendation. (visible to authors after final paper notification)

This paper presents a benchmark of 13 LLMs for designing wet-lab protocols for Tissue Optical Clearing (TOC). While the reviewers acknowledged the motivation and clarity of the benchmark setting, several weaknesses identified during the first round of reviews remain unanswered or insufficiently justified. In particular, i) the limited number of protocols and domains used for the benchmark; ii) the limited details on how the benchmark is built (e.g., the choice of prompts). These two concerns, accompanied by the weak reproducibility of the code shared by the authors, raise concerns and are major red flags for such benchmark papers. Therefore, I recommend rejection.
